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A Cross-Model Evaluation of LLM Output Reconstruction from Token-Length Traces in Controlled SSE Streaming

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† A companion IEEE TIFS draft (`tifs_main.tex`) reports bootstrap confidence intervals, paired significance tests, and a defense evaluation on the same dataset.

Abstract

Large-language-model (LLM) assistants frequently stream responses token by token using Server-Sent Events (SSE). Observable frame sizes can correlate with token lengths and thereby leak information about the generated text. This paper reports a *controlled* cross-model evaluation of output reconstruction from *idealized* one-token-per-frame SSE length traces across seven open-weight language models (six instruction/chat-tuned models and one base model) under a standardized protocol (300 aligned prompts per model; 2100 samples). A measurement harness enforces a one-token-per-frame mapping and records per-token lengths; a shared Weiss-style T5 pipeline reconstructs text hypotheses. Semantic similarity ϕ is recomputed with a single fixed term-frequency cosine backend for every sample. Mean ϕ ranges from 0.324 (Qwen2.5-1.5B-Instruct) to 0.024 (Phi-3.5-mini-instruct). We interpret the ranking as transferability of a fixed reconstructor onto each model's length traces—not as an intrinsic leakage quantity independent of tokenizer and style—and we report response-length covariates under a shared `max_new_tokens` budget. The study deliberately does *not* claim a full remote TLS traffic attack; bridging idealized length oracles to encrypted-record extraction remains future work. The results show that token-granular streaming metadata is informative enough to support model-dependent reconstruction under controlled observation.

Keywords: large language models; side-channel analysis; server-sent events; traffic analysis; privacy leakage; token streaming; text reconstruction; AI assistants

1. Introduction

Interactive LLM assistants commonly return answers incrementally to reduce perceived latency. A frequent implementation pattern is token streaming over Server-Sent Events (SSE), in which each emitted chunk is encapsulated in a text frame transmitted over an encrypted transport. While encryption protects payload content, it does not necessarily eliminate metadata leakage: observable sizes can correlate with token lengths and thereby reveal information about the generated response [1].

Weiss *et al.* demonstrated token-length side-channel reconstruction on real browser and API traffic, recovering 27% of responses and inferring the topic for 53% under their criteria [1]. Concurrent work studies merged/batched API outputs [?] and related streaming metadata attacks [?]. This paper asks a complementary question: under a controlled, idealized one-token-per-frame observation model, how does reconstruction

Received:
Accepted:
Published:

https://doi.org/10.3390/1010000

quality vary across open-weight models when prompts, decoding parameters, reconstructor, and semantic metrics are held fixed?

Scope. We study an *ideal token-length oracle* from an instrumented SSE harness, not a full remote attack on ciphertext records. An on-path observer of TLS/HTTP/2/QUIC traffic typically sees records that may coalesce or fragment SSE frames, add variable overhead, or pass through proxies. Framing the study as a controlled benchmark isolates model-dependent length patterns from transport effects; end-to-end remote extraction is future work.

The contribution of this paper is fourfold. First, we describe a controlled methodology for collecting idealized per-token length traces and reconstructing text from them. Second, we report a comparative evaluation across seven open-weight models with a single fixed semantic backend. Third, we report response-length covariates and interpret the ranking as reconstructor transfer rather than intrinsic "leakiness." Fourth, we release reproducibility artifacts; a companion draft adds confidence intervals, paired tests, and defense measurements on the same data.

2. Related Work

Side-channel attacks infer sensitive information from indirect observables such as timing, size, power, or electromagnetic emanations [2]. Network-observable metadata has long been shown to be dangerous [**? ?**]. Weiss *et al.* introduced token-length side-channel reconstruction against AI assistants, including known-plaintext fine-tuning to the victim's writing style, and evaluated the attack on live ChatGPT and Copilot traffic [1]. Li *et al.* study reconstruction when APIs merge multiple tokens into single packets [**?**], raising the novelty bar for length-channel work targeting realistic deployments. Related analyses of streaming metadata further demonstrate that encryption alone is insufficient [**?**]. Our contribution is a controlled multi-model benchmark under idealized one-token-per-frame traces rather than a new end-to-end remote exploit.

The reconstruction pipeline uses T5-style text-to-text transformers [3]. Evaluation uses normalized Levenshtein distance [4], ROUGE [5], and a fixed term-frequency cosine similarity ϕ (cf. Sentence-BERT [6]).

3. Materials and Methods

3.1. Threat Model and Experimental Scope

We assume an evaluation pipeline that obtains a sequence of per-token UTF-8 lengths and attempts to reconstruct the assistant response. In our harness, those lengths are derived from SSE frames of the form `data: <token>\n\n` via $L_t^{\mathrm{tok}} = L_t^{\mathrm{frame}} - 8$, discarding the terminal `[DONE]` frame. The reconstruction stage is fed length sequences only.

This matches the token-length channel abstraction of Weiss *et al.* [1], but measurements come from an instrumented server that *enforces* one token per SSE frame. The experiment is therefore a controlled reconstruction study under an ideal length oracle. A network adversary would still need to extract comparable lengths from ciphertext records that may merge/split frames, add variable overhead, or carry JSON wrappers—conditions we do not claim to have reproduced.

3.2. Streaming Harness and Trace Collection

The harness implements SSE streaming in a strict one-token-per-frame configuration. For every emitted token it logs token text, token UTF-8 length, SSE frame UTF-8 length, and relative emission timestamp. Token text is logged for auditing and offline metric recomputation; the attack itself uses lengths only.

$$L_t^{\mathrm{tok}} = L_t^{\mathrm{frame}} - 8, \tag{1}$$

where 8 bytes are the fixed SSE overhead (`data:` and `\n\n`).

3.3. Reconstruction Pipeline

The inversion pipeline follows a Weiss-style procedure: heuristic segmentation into sentence-like chunks; first-chunk decoding with `royweiss1/T5_FirstSentences`; subsequent chunks with `royweiss1/T5_MiddleSentences`; candidate ranking by the sum of transition log-probabilities (not length-normalized). We use the *same* public reconstructor checkpoints for every victim and do *not* fine-tune on victim-style known plaintext. Cross-model differences therefore measure *transfer* of this fixed reconstructor onto each model's length traces (tokenizer, whitespace, style, refusals, length distributions), not intrinsic leakage after per-model adaptation [1].

3.4. Models, Prompt Corpus, and Protocol

We evaluate seven open-weight language models: six instruction/chat-tuned models (Qwen2.5-1.5B/3B-Instruct, Llama-3.2-3B-Instruct, Gemma-2-2B-it, TinyLlama-1.1B-Chat, Phi-3.5-mini-instruct) and one base model (Phi-1.5), which is not instruction-tuned or RLHF-aligned according to its model card.

The prompt corpus comprises 15 topical files with 20 prompts each (300 prompts). The same prompts are issued to every model in the same order. Victim decoding is deterministic (`temperature`= 0.0, `top_p`= 1.0). The standardized subset fixes $N = 300$, `max_new_tokens`= 96, `samples_per_segment`= 3, `max_sentences`= 3, and `num_first_candidates`= 3. A shared token budget does not imply comparable amounts of text across tokenizers.

3.5. Evaluation Metrics

Reconstruction quality uses normalized Levenshtein distance ed_{norm} (↓), ROUGE-1/ROUGE-L precision (↑), and cosine semantic similarity ϕ (↑). We report the semantic recovery rate SR@0.5 (↑) as the proportion of samples with $\phi > 0.5$ (avoiding the ambiguous acronym ASR). The threshold 0.5 is a conventional operating point, not a human-calibrated criterion.

A prior mixed campaign computed ϕ with Sentence-BERT when available and TF cosine otherwise. We recompute ϕ for all 2100 samples with one fixed term-frequency cosine backend so that rankings use a comparable scale.

4. Results

4.1. Control Covariates

Under `max_new_tokens`= 96, most models truncate at or near the budget (truncation often 95–100%). Mean UTF-8 bytes per token differ substantially across models (approximately 5.0 for Qwen2.5 versus approximately 3.5 for TinyLlama and Phi-3.5-mini). Equal token counts therefore do not imply equal textual content; these covariates confound any cross-model ranking of reconstruction quality.

4.2. Cross-Model Comparison under Standardized Conditions

Table 1 reports the primary comparable subset with recomputed ϕ. Qwen2.5-1.5B-Instruct achieves the highest mean semantic similarity ($\phi = 0.3242$), followed by Qwen2.5-3B-Instruct ($\phi = 0.3096$) and Phi-1.5 ($\phi = 0.2747$ under the fixed backend). Llama-3.2-3B-

Instruct shows intermediate reconstructability ($\phi = 0.2538$), Gemma-2-2B-it $\phi = 0.1798$, TinyLlama-1.1B-Chat $\phi = 0.0428$, and Phi-3.5-mini-instruct the lowest ($\phi = 0.0239$). Only Qwen2.5-1.5B-Instruct exceeds an 8% SR@0.5. We emphasize that this ordering is the transfer ranking of one shared reconstructor.

Table 1. Primary cross-model comparison for the standardized subset ($N = 300$, `max_new_tokens`=96). ϕ (↑) recomputed with a fixed TF cosine backend; ed_{norm} (↓); ROUGE precision (↑); SR@0.5 (↑).

Model	Params	Kind	ϕ_{mean} ↑	ed_{norm} ↓	SR@0.5
Qwen2.5-1.5B-Instruct	∼1.5B	instruct	0.3242	0.7199	8.67%
Qwen2.5-3B-Instruct	∼3B	instruct	0.3096	0.7300	3.00%
Phi-1.5	∼1.3B	base	0.2747	0.7426	4.33%
Llama-3.2-3B-Instruct	∼3B	instruct	0.2538	0.7499	3.00%
Gemma-2-2B-it	∼2B	instruct	0.1798	0.7764	0.00%
TinyLlama-1.1B-Chat	∼1.1B	chat	0.0428	0.8026	0.00%
Phi-3.5-mini-instruct	∼3.8B	instruct	0.0239	0.8087	0.00%

4.3. Additional Evaluation Models

Llama-3.2-3B-Instruct, Gemma-2-2B-it, and Phi-3.5-mini-instruct were included as additional evaluation models spanning distinct families. Under fully completed 300/300 runs, relative reconstructability among them is Llama-3.2-3B > Gemma-2-2B > Phi-3.5-mini-instruct.

4.4. Topic Sensitivity

Table 2 reports best- and worst-performing topics by mean ϕ for three representative models (20 prompts per topic). Topic sensitivity exists even within a fixed attack pipeline: for Llama-3.2-3B the best topic was *employment* and the worst *safety_security*; Gemma-2-2B peaked on *beliefs_values* and was weakest on *financial_status*.

Table 2. Topic-level sensitivity based on mean semantic similarity (recomputed ϕ).

Model	Best Topic	ϕ_{mean}	Worst Topic	ϕ_{mean}
Llama-3.2-3B	employment	0.3373	safety_security	0.2053
Gemma-2-2B	beliefs_values	0.2090	financial_status	0.1575
Phi-3.5-mini	financial_status	0.0444	personal_identity	0.0073

5. Discussion

5.1. Interpretation of Reconstruction Differences

Under idealized length traces and a shared reconstructor, reconstruction quality is strongly model dependent. A service that exposes token-granular streaming lengths can enable model-dependent recovery *if* an adversary obtains those lengths. We do not claim a mechanistic explanation: tokenizer behaviour, output style, truncation under a shared budget, and reconstructor training distribution are entangled.

5.2. Reconstructor Transfer vs. Intrinsic Leakage

The ranking should be read as transferability of `royweiss1` checkpoints onto each model's traces. Weiss *et al.* improved reconstruction by fine-tuning to victim style [1]. A stronger matrix—shared reconstructor, per-model fine-tuned reconstructors, and cross-model transfer—would separate leakiness of length patterns from compatibility with a particular inversion model.

5.3. Limitations

(1) Idealized length oracle, not TLS record extraction. (2) Deterministic victim decoding. (3) Fixed public reconstructor without per-model adaptation. (4) TF cosine ϕ rather than Sentence-BERT/BERTScore. (5) SR@0.5 is conventional, not human-calibrated. (6) High truncation under `max_new_tokens`$=$ 96. Companion analyses add confidence intervals, paired Wilcoxon tests with Holm correction, and a defense sweep (bucketing, fixed framing, batching, randomized padding); those results are reported in the TIFS draft.

5.4. Ethical Considerations

This work is defensive security research: quantify reconstructability under controlled conditions to motivate safer streaming. Live offensive tooling against third-party services is out of scope.

6. Conclusions

We presented a controlled cross-model evaluation of LLM output reconstruction from idealized SSE token-length traces. Reconstruction quality is strongly model dependent under a shared reconstructor, with Qwen2.5 variants most reconstructable and Phi-3.5-mini-instruct least. Token-granular length metadata is informative under an ideal oracle; encrypted transport alone does not eliminate that channel when lengths are exposed at token granularity. The most important next steps are live encrypted-traffic extraction, per-model reconstructor adaptation, human-calibrated semantic thresholds, and production-facing defense latency measurements.

Appendix G Artifact-Completeness Rules

A run enters comparative analyses only if required files are present and internal sample counts match. Required files: `config.json`, `progress.json`, `samples.jsonl`, `summary.json`, `summary.md`, and `summary_by_topic.csv`. Incomplete placeholders are excluded from primary cross-model claims. Run-directory names belong in the reproducibility package, not in the main result tables.

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